

The Gauss-Manin connection via the de Rham space

EMERSON HEMLEY

ABSTRACT. Let k be a field of characteristic 0 and $X \rightarrow S$ be a smooth morphism of smooth k -schemes. It is a classical result that the relative algebraic de Rham cohomology $H_{\mathrm{dR}}^i(X/S)$ is naturally endowed with a flat connection, the *Gauss-Manin* connection. In this note, we provide an alternative perspective coming from the de Rham space. In particular, we explain how the relative de Rham cohomology (considered as a $\mathcal{D}_S$ -module under the Gauss-Manin connection) is isomorphic to the higher direct image of $\mathcal{O}_{X_{\mathrm{dR}}}$ along the induced morphism between de Rham spaces $X_{\mathrm{dR}} \rightarrow S_{\mathrm{dR}}$.

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1. THE DE RHAM SPACE

1.1. Preliminaries. We introduce the de Rham space, originally defined in [Sim96]. Many of the technical arguments in this section are found in [Rib24].

Let k be a fixed field of characteristic 0 and Aff_k be the category of affine k -schemes. Let τ be a Grothendieck topology on Aff_k . We define a τ algebraic stack to be a sheaf $\mathrm{Aff}_k \rightarrow \mathrm{Grpd}$ in the τ topology. The category of τ algebraic stacks will be denoted AlgStk_τ .

Definition 1.1 (de Rham space). Given any presheaf of $X : \mathrm{Aff}_k \rightarrow \mathrm{Set}$, we define X_{dR} to be the presheaf

$$X_{\mathrm{dR}}(A) := X(A^{\mathrm{red}})$$

Note that there is a natural map $X \rightarrow X_{\mathrm{dR}}$ and that associated to a morphism $X \rightarrow S$ there is an induced morphism $X_{\mathrm{dR}} \rightarrow S_{\mathrm{dR}}$. In this case, we define the relative de Rham space $(X/S)_{\mathrm{dR}}$ to be the fibered product $X_{\mathrm{dR}} \times_{S_{\mathrm{dR}}} S$.

Let X be a scheme which is formally smooth over k . If a k -algebra A has nilpotent nilradical (for instance, if A is Noetherian), then every A^{red} -point of X

extends to an A -point,

$$\begin{array}{ccc} \mathrm{Spec} A^{\mathrm{red}} & \longrightarrow & X \\ \downarrow & \nearrow \text{dashed} & \downarrow \\ \mathrm{Spec} A & \longrightarrow & \mathrm{Spec} k \end{array}$$

by infinitesimal lifting. Consequently, the natural map $X \rightarrow X_{\mathrm{dR}}$ is an epimorphism of presheaves. In general though, X smooth over k is sufficient.

Proposition 1.2. *If X is smooth over k , then $X \rightarrow X_{\mathrm{dR}}$ is an epimorphism.*

Proof. Let I_n be the ideal of A generated by elements of the form $x^n = 0$, and define $A_n := A/I_n$. Then $A^{\mathrm{red}} \cong \varinjlim_n A_n$, and

$$\mathrm{Hom}(\mathrm{Spec} A^{\mathrm{red}}, X) \cong \mathrm{Hom}(\varinjlim_n \mathrm{Spec} A_n, X) \cong \varinjlim_n \mathrm{Hom}(\mathrm{Spec} A_n, X)$$

where the second isomorphism holds because X is locally finite presentation [Sta18] Tag 01ZC. Consider an element on the right; this is given by a representative $\mathrm{Spec} A_n \rightarrow X$ for some n . By formal smoothness, this lifts to $\mathrm{Spec} A \rightarrow X$. $\square$

In view of the above proof, we also have the following characterization: if X is locally of finite presentation over k , then

$$X_{\mathrm{dR}}(A) \cong \mathrm{Hom}(\varinjlim_n \mathrm{Spec} A_n, X) \cong \varinjlim_n \mathrm{Hom}(\mathrm{Spec} A_n, X)$$

and therefore X_{dR} is a colimit. An important consequence of this is the following.

Corollary 1.3. *If X is locally of finite presentation over k , $X_{\mathrm{dR}} : \mathrm{Aff}_k \rightarrow \mathrm{Set}$ commutes with finite limits and arbitrary colimits.*

An important property of the de Rham space is that it satisfies descent.

Proposition 1.4 ([Rib24], proposition 1.1.14). *If X is a locally finite type scheme, the de Rham space X_{dR} satisfies fppf descent.*

We emphasize that the assumption k is of characteristic 0 is necessary. Thus by our conventions, X_{dR} is an fppf algebraic stack.

1.2. Algebraic de Rham cohomology. A key feature of the de Rham space is that the cover $X \rightarrow X_{\mathrm{dR}}$ provides an identification of the algebraic de Rham cohomology of X with the $\mathcal{O}$ -coherent cohomology of X_{dR} . We will provide a proof a relative version of this theorem.

Let $f : X \rightarrow S$ be a morphism of k -schemes, and consider the induced morphisms

$$\begin{array}{ccc} (X/S)_{\mathrm{dR}} & \longrightarrow & X_{\mathrm{dR}} \\ \tilde{f}_{\mathrm{dR}} \downarrow & & \downarrow f_{\mathrm{dR}} \\ S & \longrightarrow & S_{\mathrm{dR}} \end{array}$$

Let $\mathcal{O}$ be the structure sheaf of $(X/S)_{\mathrm{dR}}$.

Theorem 1.5. *If $f : X \rightarrow S$ is a morphism of smooth k -schemes,*

$$H_{\mathrm{dR}}^i(X/S) \cong R^i \tilde{f}_{\mathrm{dR}*} \mathcal{O}$$

where the left is the Zariski hypercohomology of $\Omega_{X/S}^\bullet$.

In order to prove 1.5, we need the following lemmas.

Lemma 1.6. *Let X be locally of finite presentation. Then, the formal completion of X along a closed subscheme Z may be identified as*

$$\hat{X}_Z \cong X \times_{X_{\mathrm{dR}}} Z_{\mathrm{dR}}$$

Lemma 1.7. *If X and S are locally of finite presentation,*

$$X \times_{(X/S)_{\mathrm{dR}}} X \cong (\widehat{X \times_S X})_\Delta$$

where the right is the formal completion of the image of the diagonal $X \rightarrow X \times_S X$.

Some remarks are in order. A priori, the definition of formal completion of X along a subscheme Z requires Z to be closed. However in 1.6, note that the left hand side does not require $Z \rightarrow X$ to be a closed immersion. Thus in 1.7, we do not require $X \rightarrow S$ to be separated and instead define formal completion by the above formula.

Proof. Let $Y = X \times_S X$, so that $\hat{Y}_\Delta \cong Y \times_{Y_{\mathrm{dR}}} X_{\mathrm{dR}}$. Then the fact follows from

$$\begin{array}{ccc} X_{\mathrm{dR}} \times_{S_{\mathrm{dR}}} S & \longrightarrow & Y = X \times_S X \\ \downarrow & & \downarrow \\ X_{\mathrm{dR}} & \xrightarrow{\Delta} & Y_{\mathrm{dR}} \cong X_{\mathrm{dR}} \times_{S_{\mathrm{dR}}} X_{\mathrm{dR}} \end{array}$$

being cartesian. □

Proof of 1.5. For simplicity, we consider the absolute case $S = \mathrm{Spec} k$ first. Since $\widehat{X \times X} \rightrightarrows X \rightarrow X_{\mathrm{dR}}$ is a coequalizer of presheaves, $X \rightarrow X_{\mathrm{dR}}$ is an effective epimorphism. Therefore the simplicial scheme $\hat{X}^\bullet = \check{\mathrm{Cech}}(X \rightarrow X_{\mathrm{dR}})$ is a hypercover of X_{dR} . By 1.7, $\hat{X}^n$ is the ind-scheme given by formal completion of the n -fold product $X \times \cdots \times X$ along the diagonal. By [Rod25], there is a quasi-isomorphism

$$\varprojlim \mathcal{O}_{\hat{X}^\bullet} \rightarrow \Omega_{X/k}^\bullet$$

of Zariski sheaves on X , and consequently

$$R\Gamma(X_{\mathrm{dR}}, \mathcal{O}_{X_{\mathrm{dR}}}) \cong R\Gamma(\hat{X}^\bullet, \mathcal{O}_{\hat{X}^\bullet}) \cong R\Gamma(X, \Omega_{X/k}^\bullet)$$

finishes the proof. The relative case $X \rightarrow S$ follows the same way. □

1.3. Quasi-coherent sheaves on de Rham spaces. A key insight of Simpson was that for a smooth k -scheme X , that the category of quasi-coherent sheaves over X_{dR} identifies with the category of algebraic $\mathcal{D}$ -modules on X . By 1.7 and 1.6,

$$\widehat{X \times X} \rightrightarrows X \rightarrow X_{\mathrm{dR}}$$

is a coequalizer of presheaves and thus we can view X_{dR} as a quotient of X by the relation of *infinitesimal closeness* (c.f [Lur]). The data of a quasi-coherent sheaf on X_{dR} is thus equivalent to a quasi-coherent sheaf on X along with a compatible system of isomorphism over points which lie in some n -fold formal thickening of the diagonal. This is equivalent to an action of the ring of differential operators on X .

Theorem 1.8. *If X is a smooth k -scheme, the category of quasi-coherent sheaves on X_{dR} is equivalent to the category of quasi-coherent $\mathcal{D}_X$ -modules. That is,*

$$\mathrm{QCoh}(X_{\mathrm{dR}}) \cong \mathrm{QCoh}(\mathcal{D}_X)$$

See also [Rod25] §9. In particular, a locally free sheaf on X_{dR} is equivalent to a locally free sheaf along with a flat connection on X . Pulling back a locally free sheaf along $X \rightarrow X_{\mathrm{dR}}$ corresponds to forgetting the flat connection. Therefore, the fact that the relative de Rham cohomology sheaf $H_{\mathrm{dR}}^i(X/S)$ may be endowed with a flat connection must correspond to the fact that it is pulled back from some quasi-coherent sheaf on S_{dR} .

Consider the following diagram,

$$\begin{array}{ccc} (X/S)_{\mathrm{dR}} & \xrightarrow{\tilde{\pi}} & X_{\mathrm{dR}} \\ \tilde{f}_{\mathrm{dR}} \downarrow & & \downarrow f_{\mathrm{dR}} \\ S & \xrightarrow{\pi} & S_{\mathrm{dR}} \end{array} \quad (*)$$

1.5 asserts that the relative de Rham cohomology is isomorphic to $R^i \tilde{f}_{\mathrm{dR}*} \mathcal{O}$. Recall that there is a natural map

$$\pi^* R^i f_{\mathrm{dR}*} \mathcal{O}_{X_{\mathrm{dR}}} \rightarrow R^i \tilde{f}_{\mathrm{dR}*} \tilde{\pi}^* \mathcal{O}_{X_{\mathrm{dR}}}$$

called the base-change morphism. If this map is an isomorphism, observe that this would formally endow the relative de Rham cohomology with a flat connection: since $R^i \tilde{f}_{\mathrm{dR}*} \mathcal{O}_{X_{\mathrm{dR}}}$ is a quasi-coherent sheaf on S_{dR} , it is equipped with one by 1.8. The classical construction of the Gauss-Manin connection occurs because we were going “the wrong way” around the above cartesian square.

Proposition 1.9. *Let $f : X \rightarrow S$ be a smooth morphism of smooth k -schemes. If (*) satisfies base-change, there exists a canonical flat connection ∇ on $H_{\mathrm{dR}}^i(X/S)$.*

2. THE SIX FUNCTOR FORMALISM FOR X_{dR}

Because of the non-algebraic nature of S_{dR} , establishing the above base change is slightly delicate; since S_{dR} is not a scheme, we cannot apply traditional proper or

smooth base change for quasi-coherent sheaves on schemes. To this end, we proceed by setting up some abstract machinery: we appeal to the six functor formalism for quasi-coherent sheaves on X_{dR} as found in [Rib24] or [Rod25].

2.1. Abstract 6-functor formalisms. We recall the definition of a 6-functor formalism. Rather than strive for an exhaustive treatment, we will introduce only the minimal amount of theory to state the desired result. For details see [Sch22].

Definition 2.1. A *geometric set up* a pair $(\mathcal{C}, E)$ where $\mathcal{C}$ is an ∞ -category with finite limits and E is a collection of morphisms in $\mathcal{C}$ containing all isomorphisms and stable under pull back, composition, and taking diagonals.

Remark 2.2. E is the class of $!$ -able morphisms, or morphisms which admit a $f_!$.

Definition 2.3. Given a geometric set up $(\mathcal{C}, E)$, the category of correspondences $\text{Corr}(\mathcal{C}, E)$ is defined to be the category whose objects are the objects of $\mathcal{C}$ and a morphism from X to Y is a diagram

$$\begin{array}{ccc} & Z & \\ \swarrow & & \searrow \\ X & & Y \end{array}$$

in $\mathcal{C}$ where the arrow on the right $Z \rightarrow Y$ lies in E . A composition of morphisms is

$$\begin{array}{ccccc} & & Z \times_Y S & & \\ & \swarrow & & \searrow & \\ & Z & & S & \\ \swarrow & & \searrow & & \swarrow \\ X & & Y & & T \end{array}$$

where the left arrow $Z \times_Y S \rightarrow T$ lies in E by assumption. There is a monoidal structure on $\text{Corr}(\mathcal{C}, E)$ given by the product in $\mathcal{C}$.

Definition 2.4. Given a geometric set up $(\mathcal{C}, E)$, a 3-functor formalism is a lax symmetric monoidal functor $\mathcal{D} : \text{Corr}(\mathcal{C}, E) \rightarrow \text{Cat}_{\infty}$.

Given an arbitrary morphism $f : X \rightarrow Y$ in $\mathcal{C}$, the image of the correspondence

$$\begin{array}{ccc} & X & \\ \swarrow f & & \searrow \text{id} \\ Y & & X \end{array}$$

is defined to be $f^* : \mathcal{D}(Y) \rightarrow \mathcal{D}(X)$. If $f : X \rightarrow Y$ lies in E , the image of the correspondence

$$\begin{array}{ccc} & X & \\ \swarrow \text{id} & & \searrow f \\ X & & Y \end{array}$$

is defined to be $f_! : \mathcal{D}(X) \rightarrow \mathcal{D}(Y)$.

Definition 2.5. A 6-functor formalism is a 3-functor formalism

$$\mathcal{D} : \text{Corr}(\mathcal{C}, E) \rightarrow \text{Cat}_\infty$$

such that the following holds,

- (a) the functors f^* and $f_!$ both admit right adjoints (defined to be f_* and $f^!$),
- (b) for each object of $\mathcal{C}$, the symmetric monoidal ∞ -category $\mathcal{D}(X)$ is closed.

A formal consequence of this is the following: given any Cartesian diagram

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ f' \downarrow & & \downarrow f \\ Y' & \xrightarrow{g} & Y \end{array}$$

in $\mathcal{C}$ where f is $!$ -able, there is a specified isomorphism of functors

$$g^* \circ f_! \rightarrow f'_! \circ (g')^*$$

[Man22]. This is a salient feature of 6-functor formalisms and the motivation for our consideration of them.

Definition 2.6. Let $f : X \rightarrow Y$ be $!$ -able and $\omega_f := f^!(1_Y)$ be the dualizing object. The morphism f is said to be *cohomologically smooth* provided that

- (a) The natural transformation $\omega_f \otimes f^*(-) \rightarrow f^!(-)$ is an equivalence,
- (b) ω_f is $\otimes$ -invertible in $\mathcal{D}(X)$,
- (c) For any morphism $g : Y' \rightarrow Y$ with base change $f' : X' \rightarrow Y'$ of f , the above properties also hold for f' . Moreover, if $g' : Y' \rightarrow Y$ denotes the base change of g , we also require the natural map

$$(g')^* \omega_f \rightarrow \omega_{f'}$$

to be an isomorphism.

For more details see [Sch22] §5. Cohomological smoothness is important because it yields abstract notion of smooth base change ([Zav23], lemma 2.3.9).

Theorem 2.7 (Cohomologically smooth base change). *Let*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ f' \downarrow & & \downarrow f \\ Y' & \xrightarrow{g} & Y \end{array}$$

be a Cartesian square in $\mathcal{C}$. Then the natural morphism

$$g^* \circ f_* \rightarrow (f')_* \circ (g')^*$$

is an isomorphism if g is cohomologically smooth.

As a final point, we mention briefly a particular type of cohomologically smooth morphism called cohomological étale morphisms. Intuitively, cohomological étale morphisms are cohomologically smooth with trivial dualizing object, i.e. satisfying $f^* \cong f^!$. We do not include a precise definition since cohomological smoothness is a sufficient condition for our needs (c.f. [Zav23] for a precise definition).

2.2. Quasi-coherent sheaves on algebraic stacks. We outline a construction for a 6-functor formalism for the derived category of quasi-coherent sheaves on algebraic stacks. This relies on subtle extension theorems for 6-functor formalisms which we will treat as a blackbox. This is done in full detail in [Rod25] and [Rib24].

As before, let Aff_k be the category of affine k -schemes.

Theorem 2.8. *The functor $D_{\text{qc}} : \text{Aff}_k^{\text{op}} \rightarrow \text{Cat}_{\infty}$ which maps X to its derived ∞ -category of quasi-coherent sheaves $D_{\text{qc}}(X)$ extends to a 6-functor formalism on the geometric setup $(\text{Aff}_k, \text{all})$ consisting of all morphisms.*

Proof. This is proved in [Sch22] §8.3. See also [Rib24] theorem 1.3.20 or [Rod25] proposition 6.11. $\square$

To extend this to a 6 functor formalism on algebraic stacks, we appeal to general extension results of [HM24]. This uses the D_{qc} -topology, though we will not elaborate on this since we can restrict to the coarser fppf topology.

Theorem 2.9. *The six functor formalism on the geometric set up $(\text{Aff}_k, \text{all})$ extends uniquely to a six functor formalism $(\text{AlgStk}_{D_{\text{qc}}}, E)$ such that*

- (a) *Let $f : Y \rightarrow X$ be a map whose pullback to every object in Aff_k lies in E , then f lies in E .*
- (b) *Let $f : Y \rightarrow X$ be a map which is $!$ -locally on the source or target in E ; then f lies in E .*
- (c) *Every map $f : Y \rightarrow X$ in E with $X \in \text{Sch}$ is $!$ -locally on the source in E .*

Proof. [Rod25], Corollary 6.19. $\square$

We emphasize that we do not have explicit knowledge of the class E . To check that a particular morphism of algebraic stacks is $!$ -able, we need to verify (a)-(c). For an expository account, see [Sch22] *Appendix to Lecture IV: Passage to stacks*.

Remark 2.10. The above result uses the D_{qc} -topology on algebraic stacks. However this does not pose a substantial problem; the D_{qc} -topology is *finer* than the fppf topology, and [Rib24] shows that there is a unique six functor formalism on fppf algebraic stacks whose $!$ -able morphisms are precisely the morphisms whose sheafifications lie in the E above (proposition 1.3.23). That is, there is a compatible 6-functor formalism on the geometric set up $(\text{AlgStk}_{\text{fppf}}, E')$. In the sequel, we restrict to this formalism.

We need the following technical lemmas to proceed. Let X and Y be schemes.

Lemma 2.11 ([Sch22], Proposition 8.14). *If $f : X \rightarrow Y$ is smooth and proper, then f is cohomologically smooth.*

Lemma 2.12 ([Rib24], Proposition 1.3.25). *Let $Z \subset X$ be a closed subscheme, where the ideal sheaf is locally finitely generated. Then the morphism*

$$\widehat{X}_Z \rightarrow X$$

is cohomologically étale.

Proposition 2.13. *If X is locally finite type, $X \rightarrow X_{\mathrm{dR}}$ is cohomologically smooth.*

We remark that the proof of this result is essentially ([Sch22], proposition 8.23), but Scholze works in the IndCoh-formalism.

Proof. Cohomological smoothness can be verified locally on the source and target. Since this is local on the source, we can restrict to an affine X . By taking a compactification, we can restrict to a proper X . Let $Y \rightarrow X_{\mathrm{dR}}$ be an arbitrary map from a scheme. Then the following diagram is Cartesian,

$$\begin{array}{ccc} (\widehat{X \times_k Y})_Z & \longrightarrow & Y \\ \downarrow & & \downarrow \\ X & \longrightarrow & X_{\mathrm{dR}} \end{array}$$

where $Z \subset X \times_k Y$ is the graph of $Y_{\mathrm{red}} \rightarrow X$, where Z is closed because X is separated and $Y_{\mathrm{red}} \rightarrow Y$ is a closed immersion. Then $(\widehat{X \times_k Y})_Z \rightarrow Y$ is the composite of $(\widehat{X \times_k Y})_Z \rightarrow X \times_k Y$ and $X \times_k Y \rightarrow Y$. Note that the first is cohomologically smooth by 2.12 and that the second is a base change of $X \rightarrow \mathrm{Spec} k$ and is thus cohomologically smooth by 2.11. $\square$

Proposition 2.14. *Let $f : X \rightarrow Y$ be a morphism between finite type schemes. Then $f_{\mathrm{dR}} : X_{\mathrm{dR}} \rightarrow Y_{\mathrm{dR}}$ is !-able in the 6-functor formalism for quasi-coherent sheaves.*

Proof. Since having !-functors is local on the source, it suffices to verify this after precomposing with the cohomologically smooth map $X \rightarrow X_{\mathrm{dR}}$. Then $X \rightarrow Y_{\mathrm{dR}}$ is the composite of f and $Y \rightarrow Y_{\mathrm{dR}}$, which are both !-able. $\square$

2.3. The Gauss-Manin connection. Let $f : X \rightarrow S$ be a morphism of smooth k -schemes and assume that the induced map $f_{\mathrm{dR}} : X_{\mathrm{dR}} \rightarrow S_{\mathrm{dR}}$ admits !-functors (e.g. if X and S are finite type over k). Recall the diagram

$$\begin{array}{ccc} (X/S)_{\mathrm{dR}} & \xrightarrow{\tilde{\pi}} & X_{\mathrm{dR}} \\ \tilde{f} \downarrow & & \downarrow f_{\mathrm{dR}} \\ S & \xrightarrow{\pi} & S_{\mathrm{dR}} \end{array} \quad (*)$$

2.7 and 2.13 implies the following:

Corollary 2.15. *One has base-change for diagram (*). In particular,*

$$\pi^* R^i f_{\mathrm{dR}*} \mathcal{O}_{X_{\mathrm{dR}}} \rightarrow R^i \tilde{f}_* \tilde{\pi}^* \mathcal{O}_{X_{\mathrm{dR}}}$$

is an isomorphism.

We emphasize the smoothness of f is unnecessary to establish base change for the relative de Rham square; since π is cohomologically smooth we need not assume any conditions on f_{dR} (other than that it's !-able). However, if f is not smooth, then $H_{\mathrm{dR}}^i(X/S)$ will not be a locally free sheaf and thus we speak of $\mathcal{D}_S$ -module structures instead of connections.

Corollary 2.16. *There is a canonical $\mathcal{D}_S$ -module structure on the relative de Rham cohomology $H_{\mathrm{dR}}^i(X/S)$.*

Moreover this $\mathcal{D}_S$ -module is naturally isomorphic to the i -th higher direct image of the structure sheaf of X_{dR} along f_{dR} .

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